MTH309 PRACTICE PROBLEMS WEEK 5

Question 1. Let $\{f_n\}_m$ be a sequence of $\mathbb{R}$ valued Borel measurable functions defined on Ω.

(a) Show that the functions $\omega \mapsto \sup_n f_n(\omega)$ and $\omega \mapsto \inf_n f_n(\omega)$ are $\bar{\mathbb{R}}$ valued Borel measurable functions.

(b) Show that the functions $\omega \mapsto \limsup_n f_n(\omega)$ and $\omega \mapsto \liminf_n f_n(\omega)$ are $\bar{\mathbb{R}}$ valued Borel measurable functions.

Question 2. Let X be an RV defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with law $\mathbb{P} \circ X^{-1}$ and DF F_X.

(a) Show that F_X has at most countably many points of discontinuities.

(b) Suppose $D = \{x_1, x_2, \cdots\}$ denote the set of discontinuities of F_X. Recall that $F_X(x_j) - F_X(x_j-)$ is the size of the jump at x_j. Define the function $F_d : \mathbb{R} \to \mathbb{R}$ by

$$F_d(x) := \sum_{x_j \in D} (F_X(x_j) - F_X(x_j-)) 1_{(-\infty, x]}(x_j), x \in \mathbb{R},$$

which looks at contribution from all the jumps on or before any point x. Show that

(i) $0 \leq F_d(x) \leq F_X(x) \leq 1, \forall x \in \mathbb{R}$.

(ii) F_d is right-continuous and non-decreasing.

(iii) $\lim_{x \to -\infty} F_d(x)$ and $\lim_{x \to \infty} F_d(x)$ exist.

(c) Define the function $F_c : \mathbb{R} \to \mathbb{R}$ by $F_c(x) := F_X(x) - F_d(x), x \in \mathbb{R}$. Show that

(i) $0 \leq F_c(x) \leq F_X(x) \leq 1, \forall x \in \mathbb{R}$.

(ii) F_c is continuous and non-decreasing.

(iii) $\lim_{x \to -\infty} F_c(x)$ and $\lim_{x \to \infty} F_c(x)$ exist.

(d) Set $\alpha := \lim_{x \to \infty} F_d(x)$ and $\beta := \lim_{x \to \infty} F_c(x)$. Show that

(i) $0 \leq \alpha \leq 1$, $0 \leq \beta \leq 1$.

(ii) $\alpha + \beta = 1$.

(iii) If $\alpha > 0$, show that $\frac{1}{\alpha} F_d$ is a discrete DF.

(iv) If $\alpha = 1$, then F_X is discrete and $F_c \equiv 0$.

(v) If $\beta > 0$, show that $\frac{1}{\beta} F_c$ is a continuous DF.

(vi) If $\beta = 1$, then F_X is continuous and $F_d \equiv 0$.

Note: $F_X = \alpha \times \frac{1}{\alpha} F_d + \beta \times \frac{1}{\beta} F_c$ is a convex combination of DFs. Here, if $\alpha = 0$, we interpret the term $\alpha \times \frac{1}{\alpha} F_d$ as 0.

Question 3. Show that $\mathbb{B}_{\mathbb{R}} = \mathbb{B}_{\overline{\mathbb{R}}} \cap \mathbb{R}$.

Question 4. Let $\{A_n\}_n$ be an increasing sequence of sets from a σ-field $\mathcal{F}$ and write $A = \bigcup_n A_n$, i.e. $A_n \uparrow A$. Is the fact "$1_{A_n} \uparrow 1_A$" true?

Question 5. Recall the approximation of Borel measurable functions by simple functions, as discussed in the class. If the original function is bounded, show that the convergence is actually uniform.